

ODE and simple PDE. Initial and boundary conditions.

SOLUTIONS TO EXAMPLES

1. The equation

$$\frac{d^2y}{dx^2} + \lambda y = 0 \quad (1)$$

is a 2nd-order linear homogeneous differential equation with constant coefficients¹. We seek its solution in the form

$$y = e^{sx},$$

where s is a constant to be determined. Its derivatives are

$$y' = se^{sx}, \quad y'' = s^2e^{sx}.$$

Substituting the expressions for y and y'' into Eq. (1) gives

$$s^2e^{sx} + \lambda e^{sx} = 0.$$

Multiplying this equation by e^{-sx} leads to the *auxiliary equation* for s :

$$s^2 + \lambda = 0. \quad (2)$$

Solving this quadratic equation, we have

$$s^2 = -\lambda,$$

so that

$$s = \pm\sqrt{-\lambda}.$$

If $\lambda \neq 0$, the auxiliary equation has two distinct roots, $s = \sqrt{-\lambda}$ and $s = -\sqrt{-\lambda}$. The general solution of Eq. (1) is then given by the linear combination of the two exponential solutions,

$$y = C_1e^{\sqrt{-\lambda}x} + C_2e^{-\sqrt{-\lambda}x}, \quad (3)$$

where C_1 and C_2 are arbitrary coefficients.

To implement the first boundary condition, $y'(0) = 0$, we use

$$y' = \sqrt{-\lambda}C_1e^{\sqrt{-\lambda}x} - \sqrt{-\lambda}C_2e^{-\sqrt{-\lambda}x},$$

and set $x = 0$, which gives

$$\sqrt{-\lambda}(C_1 - C_2) = 0.$$

Since $\lambda \neq 0$, we have

$$C_2 = C_1 \equiv C,$$

so that

$$y = C \left(e^{\sqrt{-\lambda}x} + e^{-\sqrt{-\lambda}x} \right). \quad (4)$$

The second boundary condition, $y(l) = 0$, then gives

$$C \left(e^{\sqrt{-\lambda}l} + e^{-\sqrt{-\lambda}l} \right) = 0. \quad (5)$$

¹Such equations and their solution were considered in MTH1021.

For $\lambda < 0$ both exponents in Eq. (5) are real and positive, so the expression in brackets is also positive, and Eq. (5) gives $C = 0$, so that $y = 0$, i.e., a trivial solution.

For $\lambda > 0$, we have $\sqrt{-\lambda} = i\sqrt{\lambda}$, and Eq. (4) gives

$$y = C \left(e^{i\sqrt{\lambda}x} + e^{-i\sqrt{\lambda}x} \right). \quad (6)$$

Using Euler's identity, $e^{i\theta} = \cos \theta + i \sin \theta$, and its complex conjugate, $e^{-i\theta} = \cos \theta - i \sin \theta$, we have

$$y = 2C \cos(\sqrt{\lambda}x),$$

or, renaming $2C$ by C ,

$$y = C \cos(\sqrt{\lambda}x). \quad (7)$$

The boundary condition $y(l) = 0$ requires

$$C \cos(\sqrt{\lambda}l) = 0.$$

Recalling that $\cos \theta = 0$ for $\theta = \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$, we have

$$\sqrt{\lambda}l = \frac{\pi}{2} + \pi n, \quad (8)$$

with $n = 0, 1, 2, \dots$ (while negative n are excluded, since $\sqrt{\lambda}l > 0$.) Hence, the values of λ for which Eq. (1) with the boundary conditions $y'(0) = y(l) = 0$ has nonzero zero solutions, are found from

$$\sqrt{\lambda} = \frac{\pi \left(n + \frac{1}{2} \right)}{l},$$

which gives

$$\lambda = \frac{\pi^2 \left(n + \frac{1}{2} \right)^2}{l^2}, \quad n = 0, 1, 2, \dots \quad (9)$$

The corresponding solutions are

$$y = C \cos \left[\pi \left(n + \frac{1}{2} \right) \frac{x}{l} \right]. \quad (10)$$

The first three solutions ($n = 0, 1, 2$) are shown in figure 1 using $C = 1$.

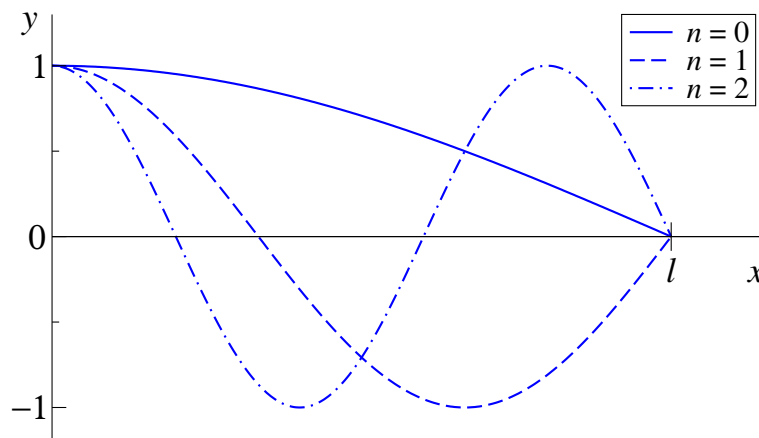


Figure 1: Non-zero solutions of Eq. (1) with boundary conditions $y'(0) = 0$, $y(l) = 0$, with λ given by Eq. (9) and $y(x)$ given by Eq. (10).

These solutions in fact describe standing waves, e.g., those that occur when air vibrates in a semiclosed (organ) pipe of length l . The function $y(x)$ represents the amplitude of the

displacement of air along the pipe, with the end at $x = 0$ being open (hence, maximum displacement) and that at $x = l$ being closed (zero displacement).

In the above we assumed that $\lambda \neq 0$. For $\lambda = 0$, Eq. (1) gives

$$\frac{d^2 y}{dx^2} = 0.$$

Integrating this once gives $dy/dx = A$, and integrating again gives

$$y = Ax + B,$$

where A and B are arbitrary constants. The condition $y'(0) = 0$ yields $A = 0$, so that $y = B$, and from $y(l) = 0$, we find $B = 0$, so that only a trivial (zero) solution is possible.

Finally, we assumed from the start that $\lambda \in \mathbb{R}$. It is not difficult to show that the boundary conditions $y'(0) = y(l) = 0$ cannot be satisfied for any complex λ with a non-zero imaginary part, i.e., the only possible values of λ are those given by Eq. (9).

2. To find the partial differential equation satisfied by all functions of the form $u = f(x^2 - y^2)$, we need to take derivatives of u with respect to x and y and manipulate the expressions obtained in such a way as to eliminate the function f .

We obtain (using the chain rule)

$$u_x \equiv \frac{\partial u}{\partial x} = f'(x^2 - y^2)2x, \quad (11)$$

and

$$u_y \equiv \frac{\partial u}{\partial y} = f'(x^2 - y^2)(-2y), \quad (12)$$

where f' denotes the derivative of f with respect to its argument.

Dividing Eq. (11) by Eq. (12) (to eliminate f'), we have

$$\frac{u_x}{u_y} = -\frac{2x}{2y},$$

which gives $yu_x = -xu_y$, so the required differential equation is

$$yu_x + xu_y = 0.$$

3. Integrating the equation

$$u_{xxx} + u_x = 0 \quad (13)$$

with respect to x , and keeping in mind that u is a function of x , y and z , we obtain

$$u_{xx} + u = f(y, z), \quad (14)$$

where $f(y, z)$ is an arbitrary function of y and z , which plays the role of an arbitrary integration constant. [Differentiation of Eq. (14) with respect to x obviously gives (13).]

Equation (14) is a linear inhomogeneous equation. Solving the homogeneous equation first,

$$u_{xx} + u = 0,$$

we have²

$$u = g(y, z) \cos x + h(y, z) \sin x, \quad (15)$$

²For a function of x only, the equation $u'' + u = 0$ has the general solution $u = C_1 \cos x + C_2 \sin x$.

where $g(y, z)$ and $h(y, z)$ are arbitrary functions (i.e., “constants” with respect to x).

To find the solution for the inhomogeneous equation (14), it is sufficient to add $f(y, z)$ to the above solution (this function playing the role of a particular integral), which yields

$$u(x, y, z) = f(y, z) + g(y, z) \cos x + h(y, z) \sin x. \quad (16)$$

One can check that this is correct by substitution into Eq. (14).

4. The string with the ends fixed at $x = 0$ and $x = l$, and mass m_0 attached at $x = a$, and tension T along the string, is shown in figure 2.

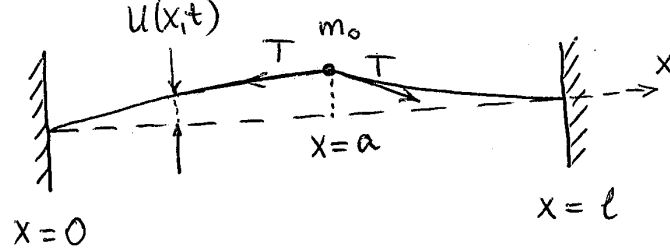


Figure 2: The shape of a string with a mass m_0 attached at $x = a$ and tension T .

The boundary conditions for the string displacement $u(x, t)$ are $u(0, t) = 0$ and $u(l, t) = 0$. For $0 \leq x < a$ and $a < x \leq l$, the function $u(x, t)$ satisfies the wave equation for a uniform string,

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0,$$

where $c = \sqrt{T/\rho}$, and ρ is the linear mass density of the string.

The net force acting on m_0 in the direction perpendicular to x is determined by the vertical components of the two tension forces

$$T \sin \theta|_{x=a+0} - T \sin \theta|_{x=a-0},$$

where θ is the angle that the string makes with the x axis, while $x = a + 0$ and $x = a - 0$ indicate that values of θ are taken “just to the right” and “just to the left” of a (i.e., one-sided limits, sometimes denoted by $x \rightarrow a^+$ and $x \rightarrow a^-$).

For small angles θ , $|\theta| \ll 1$, we can use

$$\sin \theta \simeq \tan \theta = \frac{\partial u}{\partial x}.$$

Hence, the force acting on m_0 is

$$T \left(\frac{\partial u}{\partial x} \Big|_{x=a+0} - \frac{\partial u}{\partial x} \Big|_{x=a-0} \right),$$

and Newton’s second law for m_0 reads

$$m_0 \frac{\partial^2 u}{\partial t^2} \Big|_{x=a} = T \left(\frac{\partial u}{\partial x} \Big|_{x=a+0} - \frac{\partial u}{\partial x} \Big|_{x=a-0} \right) \quad (17)$$

Therefore, in order to solve the problem, one needs to construct a solution on $0 \leq x \leq a$ that satisfies $u(0, t) = 0$; let’s denote it $u_1(x, t)$. Then one needs to do the same for $a \leq x \leq l$ with $u(l, t) = 0$; denote this solution $u_2(x, t)$. These two solutions must match at $x = a$,

$$u_1(a, t) = u_2(a, t),$$

and satisfy the condition given by Eq. (17) at $x = a$:

$$\frac{m_0}{T} \frac{\partial^2 u}{\partial t^2} \Big|_{x=a} = \frac{\partial u_2}{\partial x} \Big|_{x=a} - \frac{\partial u_1}{\partial x} \Big|_{x=a}.$$